Supplementary Material: Worse Prognosis, Larger Benefit

A Bayesian Hierarchical Ordinal Meta-analysis of 30 Thrombectomy Trials

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Search Strategy

We searched PubMed (MEDLINE via NCBI) from inception to October 19, 2025, with no date limits. The full, reproducible PubMed strategy was:

```
(
"mechanical thrombectomy"[tiab] OR "endovascular thrombectomy"[tiab] OR
"aspiration thrombectomy"[tiab] OR "stent retriever*"[tiab] OR stentriever*[tiab] OR
Solitaire[tiab] OR Trevo[tiab] OR Penumbra[tiab]
)
AND
(
stroke[tiab] OR "ischemic stroke"[tiab] OR "large vessel occlusion"[tiab] OR LVO[tiab] OR
"basilar artery"[tiab] OR "basilar occlusion"[tiab]
)
AND
(
randomized controlled trial[Publication Type] OR random*[tiab] OR trial[tiab]
)
NOT (animals[mh] NOT humans[mh])
AND english[lang]
```

This search returned 327 records.

Eligibility Criteria

- **Design:** Randomized.
- **Population:** Adults with acute ischemic stroke (anterior or posterior circulation).
- **Intervention:** Mechanical thrombectomy (including stent retrievers and/or aspiration).
- **Comparator:** Best medical management (with or without intravenous [IV] thrombolysis).
- **Outcomes:** Functional outcomes (modified Rankin Scale).
- **Exclusions:** Non-randomized studies.

Study Selection and Data Handling

We screened titles and abstracts, then full texts, with disagreements resolved by consensus, as documented in the PRISMA 2020 diagram below.

For each included trial, we extracted trial design, population, time window, imaging selection, intervention details (device/approach), comparator, and outcomes (including full-ordinal mRS where available).

For trials reporting mRS results as percentages, counts were derived by multiplying each category percentage by the total intention-to-treat sample size. This assumes each percentage was calculated from the full intention-to-treat denominator, such that counts across all six mRS categories sum to the total enrolled.

Protocol Amendments

This review was pre-registered at the Open Science Framework (https://osf.io/smcen/) in April 2023. The original pre-registration described a focused meta-analysis of three randomized trials of mechanical thrombectomy for large ischemic stroke, with a primary objective of estimating the average treatment effect using a Bayesian multilevel model and a primary outcome of mRS 0–2 at 90 days. The present study departs from that pre-registration in three respects. First, the scope expanded from large-core trials specifically to all 30 published randomized thrombectomy trials, reflecting the rapid growth of the literature between 2023 and 2025 and the recognition that the central scientific question — whether worse-prognosis populations derive larger or smaller relative benefit from thrombectomy — required the full trial record to answer. Second, the primary estimand changed from the pooled average treatment effect to the trial-level intercept–slope correlation ( ), which quantifies the prognosis-benefit gradient directly. Third, the primary outcome changed from the dichotomized mRS 0–2 endpoint to the full 90-day ordinal mRS distribution, which makes more complete use of the available data. These amendments are disclosed here in accordance with PRISMA 2020 item 24c. All analytic code and data are publicly available at the OSF repository (https://osf.io/smcen/), making every modeling decision fully transparent and reproducible; in the context of complete open-data and open-code publication, the distinction between pre-registered and emergent analyses is largely one of documentation rather than scientific integrity.

PRISMA Diagram

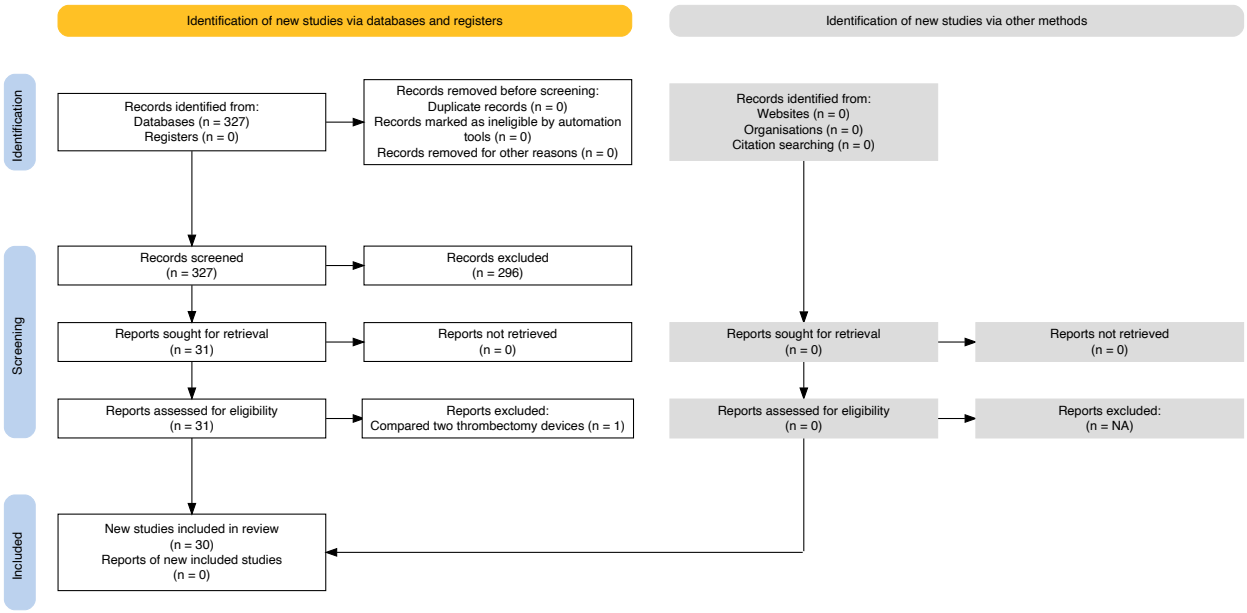

Figure 1: PRISMA 2020 diagram of search strategy and study selection.

Detailed References

All 30 included trials, listed in chronological order of publication.

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Trial Characteristics

Table 1

Trial	Year	N (EVT)	N (Ctrl)	Occlusion	Time Window	Population	Primary Endpoint	Key Feature	Notes
IMS III	2013	415	214	ICA/M1	6 h	Early-window standard LVO	mRS 0-2 90d	IVT eligible	
SYNTHESIS	2013	181	181	Mixed	4.5 h	Early-window standard LVO	mRS 0-1 90d	No CTA requirement	
MR RESCUE	2013	63	54	ICA/M1	8 h	Early-window standard LVO	mRS shift 90d	Imaging strata	
EXTEND-IA	2015	35	35	ICA/M1	6 h	Early-window standard LVO	mRS shift 90d	Perfusion-selected	
SWIFT PRIME	2015	98	93	ICA/M1	6 h	Early-window standard LVO	mRS shift 90d	IVT eligible	
ESCAPE	2015	163	146	ICA/M1	12 h	Early-window standard LVO	mRS shift 90d	Good collaterals	
REVASCAT	2015	103	103	ICA/M1	8 h	Early-window standard LVO	mRS shift 90d	ASPECTS 7	
MR CLEAN	2015	233	266	ICA/M1	6 h	Early-window standard LVO	mRS shift 90d	Pragmatic	
THERAPY	2016	50	46	ICA/M1	6 h	Early-window standard LVO	mRS shift 90d	IVT eligible	
THRACE	2016	200	202	ICA/M1	5 h	Early-window standard LVO	mRS 0-2 90d	IVT eligible	
THRILL	2016	2	2	ICA/M1	6 h	Early-window standard LVO	Not reported	IVT ineligible	Stopped early
PISTE	2017	33	30	ICA/M1	6 h	Early-window standard LVO	mRS shift 90d	Pragmatic	
EASI	2017	37	40	Mixed	6 h	Early-window standard LVO	mRS 0-2 90d	Small RCT	
DEFUSE-3	2018	92	90	ICA/M1	6-16 h	Late-window/mismatch	mRS shift 90d	Core <70 mL	
DAWN	2018	108	100	ICA/M1	6-24 h	Late-window/mismatch	mRS shift 90d	Clinical-core mismatch	Stopped early
RESILIENT	2020	111	112	ICA/M1	8 h	Early-window standard LVO	mRS 0-2 90d	Public health system	
BEST	2020	66	65	Basilar	8 h	Basilar artery occlusion	mRS 0-3 90d	BAO only	Stopped early
BASICS	2021	154	146	Basilar	6 h	Basilar artery occlusion	mRS shift 90d	BAO only	

(continued)

Trial	Year	N (EVT)	N (Ctrl)	Occlusion	Time Window	Population	Primary Endpoint	Key Feature	Notes
POSITIVE	2022	12	21	Mixed	12 h	Late-window/mismatch	mRS shift 90d	Perfusion-selected	
BAOCHE	2022	111	106	Basilar	6-24 h	Basilar artery occlusion	mRS shift 90d	NIHSS 10	
ATTENTION	2022	225	115	Basilar	12 h	Basilar artery occlusion	mRS shift 90d	BAO only	
RESCUE-Japan LIMIT	2022	100	102	ICA/M1	6 h	Large-core	mRS shift 90d	ASPECTS 3-5	
SELECT2	2023	177	171	ICA/M1	24 h	Large-core	mRS shift 90d	ASPECTS 3-5 or core 50 mL	Stopped early
MR CLEAN-LATE	2023	253	247	ICA/M1	6-24 h	Late-window	mRS shift 90d	Good collaterals	
ANGEL-ASPECT	2023	230	225	ICA/M1	24 h	Large-core	mRS shift 90d	Core 70 mL	
TENSION	2023	124	122	ICA/M1	12 h	Large-core	mRS shift 90d	ASPECTS 3-5	
TESLA	2024	151	146	ICA/M1	6 h	Large-core	mRS shift 90d	ASPECTS 5	
LASTE	2024	159	165	ICA/M1	6.5 h	Large-core	mRS shift 90d	ASPECTS 5	Stopped early
DISTAL	2025	271	269	Distal/DMVO	6 h	Medium/distal vessel	mRS shift 90d	DMVO only	
ESCAPE-MeVO	2025	255	274	M2/MeVO	12 h	Medium vessel	mRS shift 90d	MeVO only	

Risk of Bias Assessment

Table 2

Trial	Random Sequence	Allocation Concealment	Blinding (Performance)	Blinding (Detection)	Attrition	Selective Reporting	Other
TESLA	Low risk	Low risk	High risk	Low risk	Low risk	Low risk	
DISTAL	Low risk	Low risk	High risk	Low risk	Low risk	Low risk	
ESCAPE-MeVO	Low risk	Low risk	High risk	Low risk	Low risk	Low risk	
TENSION	Low risk	Low risk	High risk	Low risk	Low risk	Low risk	
THERAPY	Unclear risk	Low risk	High risk	Low risk	Unclear risk	Low risk	Low risk
MR CLEAN-LATE	Low risk	Low risk	High risk	Low risk	Low risk	Low risk	
THRACE	Low risk	Low risk	High risk	Low risk	Unclear risk	Low risk	Low risk
PISTE	Low risk	Low risk	High risk	Low risk	Low risk	Low risk	Low risk
POSITIVE	Low risk	Low risk	High risk	Low risk	Low risk	Low risk	
RESILIENT	Low risk	Low risk	High risk	Low risk	Low risk	Low risk	Low risk
DEFUSE-3	Low risk	Low risk	High risk	Low risk	Unclear risk	Low risk	Low risk
DAWN	Low risk	Low risk	High risk	Low risk	Low risk	Low risk	Low risk
BASICS	Low risk	Low risk	High risk	Low risk	Low risk	Low risk	
BEST	Low risk	Low risk	High risk	Low risk	Low risk	Low risk	Unclear risk
BAOCHE	Low risk	Low risk	High risk	Low risk	Low risk	Low risk	
ATTENTION	Low risk	Low risk	High risk	Low risk	Low risk	Low risk	
EXTEND-IA	Low risk	Low risk	High risk	Low risk	Low risk	Low risk	Unclear risk
SWIFT PRIME	Low risk	Low risk	High risk	Low risk	Low risk	Low risk	Low risk
ESCAPE	Low risk	Low risk	High risk	Low risk	Low risk	Low risk	Unclear risk
REVASCAT	Low risk	Low risk	High risk	Low risk	Low risk	Low risk	Low risk
MR CLEAN	Low risk	Low risk	High risk	Low risk	Low risk	Low risk	Low risk
IMS III	Low risk	Low risk	High risk	Low risk	Low risk	Low risk	Unclear risk
SYNTHESIS	Low risk	Low risk	Unclear risk	Low risk	Low risk	Low risk	
MR RESCUE	Low risk	Low risk	High risk	Low risk	High risk	Low risk	Low risk
RESCUE-Japan LIMIT	Low risk	Low risk	High risk	Low risk	Low risk	Low risk	
ANGEL-ASPECT	Low risk	Low risk	High risk	Low risk	Low risk	Low risk	
SELECT2	Low risk	Low risk	High risk	Low risk	Low risk	Low risk	
THRILL	Low risk	Unclear risk	High risk	Unclear risk	High risk	High risk	
EASI	Low risk	Unclear risk	High risk	High risk	Low risk	Low risk	
LASTE	Low risk	Low risk	High risk	Low risk	Low risk	Low risk	

Main Model Specification

The main model was a Bayesian hierarchical proportional-odds cumulative ordinal model fit with `brms` (Stan backend via `cmdstanr`). The model formula was:

```
 Family: cumulative
  Links: mu = logit
Formula: mrs_better | weights(count) ~ treatment + (1 + treatment | trial)
   Data: data_long %>% filter(count > 0) (Number of observations: 403)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~trial (Number of levels: 30)
                                     Estimate Est.Error l-95% CI u-95% CI Rhat
sd(Intercept)                            0.76      0.11     0.58     1.00 1.01
sd(treatmentthrombectomy)                0.35      0.08     0.22     0.51 1.00
cor(Intercept,treatmentthrombectomy)    -0.49      0.18    -0.80    -0.11 1.00
                                     Bulk_ESS Tail_ESS
sd(Intercept)                             872     1614
sd(treatmentthrombectomy)                1237     2200
cor(Intercept,treatmentthrombectomy)     2104     2785

Regression Coefficients:
                      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
Intercept[1]             -1.07      0.14    -1.36    -0.80 1.00      535
Intercept[2]             -0.51      0.14    -0.80    -0.24 1.00      530
Intercept[3]              0.27      0.14    -0.02     0.53 1.00      533
Intercept[4]              0.97      0.14     0.68     1.23 1.00      542
Intercept[5]              1.71      0.14     1.41     1.97 1.00      554
Intercept[6]              2.93      0.14     2.62     3.20 1.00      586
treatmentthrombectomy     0.46      0.08     0.30     0.62 1.00     1229
                      Tail_ESS
Intercept[1]               839
Intercept[2]               886
Intercept[3]               831
Intercept[4]               852
Intercept[5]               963
Intercept[6]               982
treatmentthrombectomy     1620

Further Distributional Parameters:
     Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
disc     1.00      0.00     1.00     1.00   NA       NA       NA

Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
```

```
scale reduction factor on split chains (at convergence, Rhat = 1).
```

All Rhat values ≤ 1.01, consistent with convergence. Bulk and tail effective sample sizes were adequate (all >400 for key parameters).

Alternative Models

Three additional models were fit to assess robustness of the main findings.

Alternative Model 1: Ordinal (PO + Unequal Variances)

Formula: `mrs_better | weights(count) ~ treatment + (1 + treatment | trial); disc ~ 0 + treatment` Family: cumulative (logit link, log disc link) Chains: 4 | Iterations: 2000 (1000 warmup) | Total post-warmup draws: 4000

```
 Family: cumulative
  Links: mu = logit; disc = log
Formula: mrs_better | weights(count) ~ treatment + (1 + treatment | trial)
         disc ~ 0 + treatment
   Data: data_long (Number of observations: 420)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~trial (Number of levels: 30)
                                     Estimate Est.Error l-95% CI u-95% CI Rhat
sd(Intercept)                            0.80      0.12     0.61     1.07 1.00
sd(treatmentthrombectomy)                0.37      0.08     0.23     0.55 1.00
cor(Intercept,treatmentthrombectomy)    -0.47      0.19    -0.78    -0.06 1.00
                                     Bulk_ESS Tail_ESS
sd(Intercept)                             793     1500
sd(treatmentthrombectomy)                1326     2374
cor(Intercept,treatmentthrombectomy)     2186     2923

Regression Coefficients:
                          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
Intercept[1]                 -1.18      0.15    -1.49    -0.89 1.01      538
Intercept[2]                 -0.58      0.15    -0.89    -0.29 1.01      535
Intercept[3]                  0.25      0.15    -0.06     0.54 1.01      533
Intercept[4]                  1.01      0.15     0.69     1.29 1.01      536
Intercept[5]                  1.81      0.15     1.49     2.10 1.01      553
Intercept[6]                  3.13      0.16     2.79     3.43 1.01      603
treatmentthrombectomy         0.48      0.09     0.31     0.65 1.00     1484
disc_treatmentthrombectomy   -0.14      0.02    -0.18    -0.09 1.00     3234
                          Tail_ESS
Intercept[1]                  1042
Intercept[2]                   968
Intercept[3]                   953
Intercept[4]                  1017
Intercept[5]                  1090
Intercept[6]                   989
treatmentthrombectomy         2206
```

```
disc_treatmentthrombectomy     2786

Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Alternative Model 2: Ordinal (Adjacent-Category)

Formula: `mrs_better | weights(count) ~ treatment + (1 + treatment | trial)` Family: acat (adjacent-category logit) Chains: 4 | Iterations: 2000 (1000 warmup) | Total post-warmup draws: 4000

```
 Family: acat
  Links: mu = logit
Formula: mrs_better | weights(count) ~ treatment + (1 + treatment | trial)
   Data: data_long %>% filter(count > 0) (Number of observations: 403)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~trial (Number of levels: 30)
                                     Estimate Est.Error l-95% CI u-95% CI Rhat
sd(Intercept)                            0.25      0.04     0.19     0.33 1.00
sd(treatmentthrombectomy)                0.11      0.02     0.07     0.17 1.00
cor(Intercept,treatmentthrombectomy)    -0.62      0.16    -0.87    -0.28 1.00
                                     Bulk_ESS Tail_ESS
sd(Intercept)                             784     1553
sd(treatmentthrombectomy)                1308     2274
cor(Intercept,treatmentthrombectomy)     2138     2466

Regression Coefficients:
                      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
Intercept[1]              0.82      0.06     0.69     0.94 1.00      783
Intercept[2]             -0.42      0.06    -0.54    -0.30 1.00      899
Intercept[3]              0.19      0.06     0.07     0.31 1.00      801
Intercept[4]              0.27      0.06     0.14     0.39 1.00      876
Intercept[5]              0.18      0.06     0.06     0.31 1.00      832
Intercept[6]              0.74      0.07     0.61     0.88 1.00     1066
treatmentthrombectomy     0.15      0.03     0.10     0.20 1.00     1044
                      Tail_ESS
Intercept[1]              1491
Intercept[2]              1773
Intercept[3]              1591
Intercept[4]              1900
Intercept[5]              1886
Intercept[6]              2077
treatmentthrombectomy     1702
```

```
Further Distributional Parameters:
     Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
disc     1.00      0.00     1.00     1.00   NA       NA       NA

Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Alternative Model 3: Binary (mRS 0–2 vs 3–6)

Formula: `good_n | trials(total_n) ~ treatment + (1 + treatment | trial)` Family: binomial (logit link) Chains: 4 | Iterations: 2000 (1000 warmup) | Total post-warmup draws: 4000

```
 Family: binomial
  Links: mu = logit
Formula: good_n | trials(total_n) ~ treatment + (1 + treatment | trial)
   Data: data_bin (Number of observations: 60)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~trial (Number of levels: 30)
                                     Estimate Est.Error l-95% CI u-95% CI Rhat
sd(Intercept)                            0.98      0.14     0.74     1.29 1.00
sd(treatmentthrombectomy)                0.54      0.10     0.37     0.78 1.01
cor(Intercept,treatmentthrombectomy)    -0.65      0.14    -0.88    -0.32 1.00
                                     Bulk_ESS Tail_ESS
sd(Intercept)                             966     1923
sd(treatmentthrombectomy)                1198     2086
cor(Intercept,treatmentthrombectomy)     2193     2636

Regression Coefficients:
                      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
Intercept                -1.16      0.18    -1.52    -0.81 1.01      556
treatmentthrombectomy     0.67      0.12     0.44     0.90 1.00     1034
                      Tail_ESS
Intercept                 1215
treatmentthrombectomy     1740

Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

All three alternative models yielded a consistently negative intercept–slope correlation, confirming robustness of the main finding.

eFigure 1: Robustness of ρ to Model Specification

Posterior medians and 95% credible intervals for ρ under the primary proportional-odds ordinal model and three alternative models: an unequal-variance proportional-odds model, an adjacent-category ordinal model, and a binary model (mRS 0–2 vs 3–6). Estimates remain negative across all specifications.

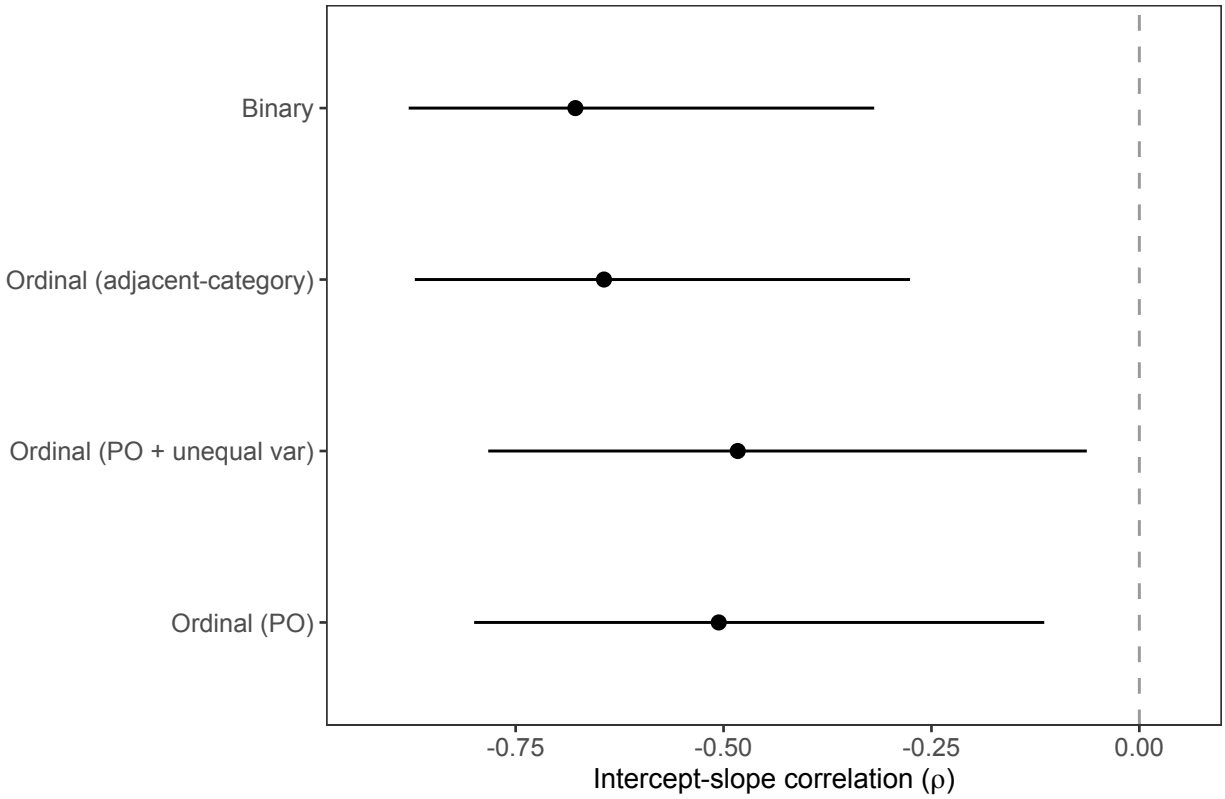

Figure 2: Robustness of the intercept–slope correlation to model specification.

eFigure 2: Absolute Benefit Across Model Specifications

For each model, points show each trial's posterior median control-group probability of functional independence (x-axis) and posterior median absolute benefit in functional independence due to thrombectomy (y-axis). Functional independence is defined as mRS 0–2. The dashed line indicates no absolute benefit; the smooth curve is a descriptive fit with 95% uncertainty band.

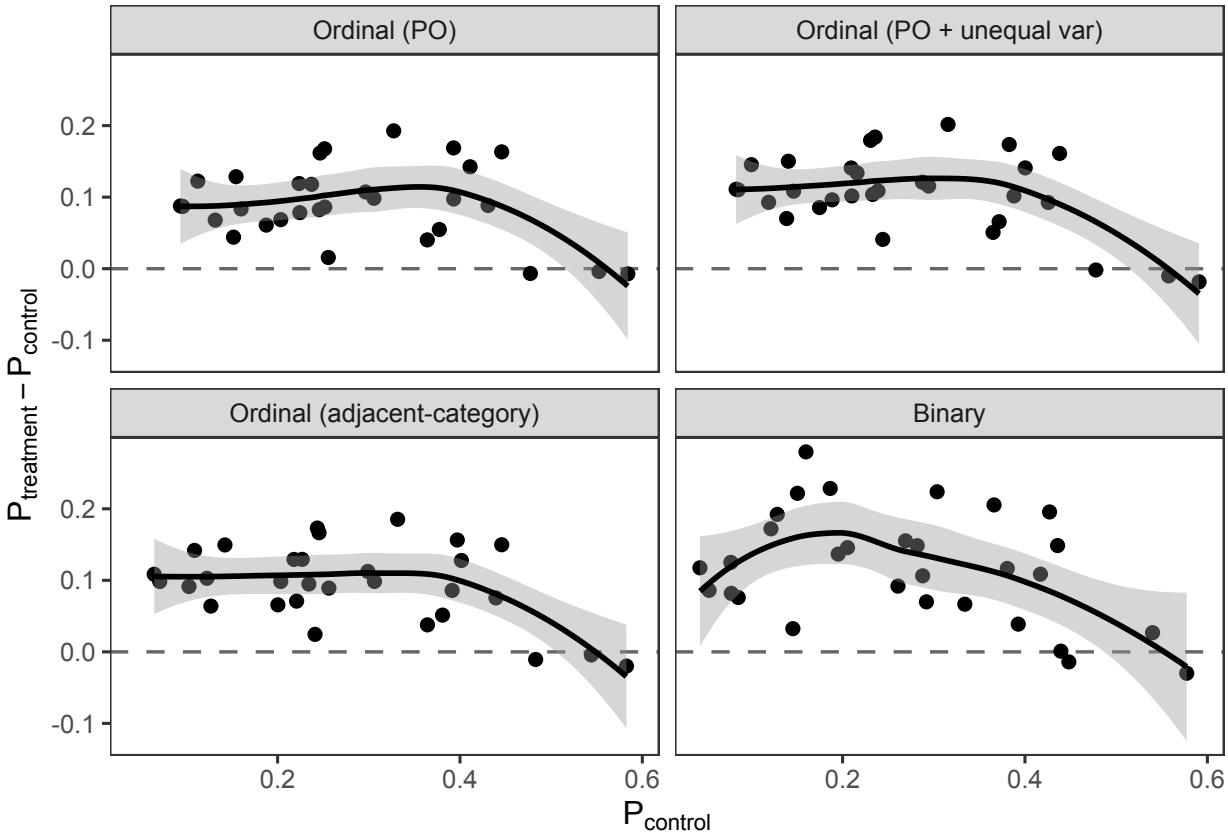

Figure 3: Baseline prognosis and absolute benefit in functional independence.

Model Diagnostics

The following diagnostic figures are generated by running `run.R` and are saved to `master_supporting_docs/support`

Trace Plots

Markov chain trace plots for the six threshold parameters, the treatment effect, and the three variance-covariance hyperparameters.

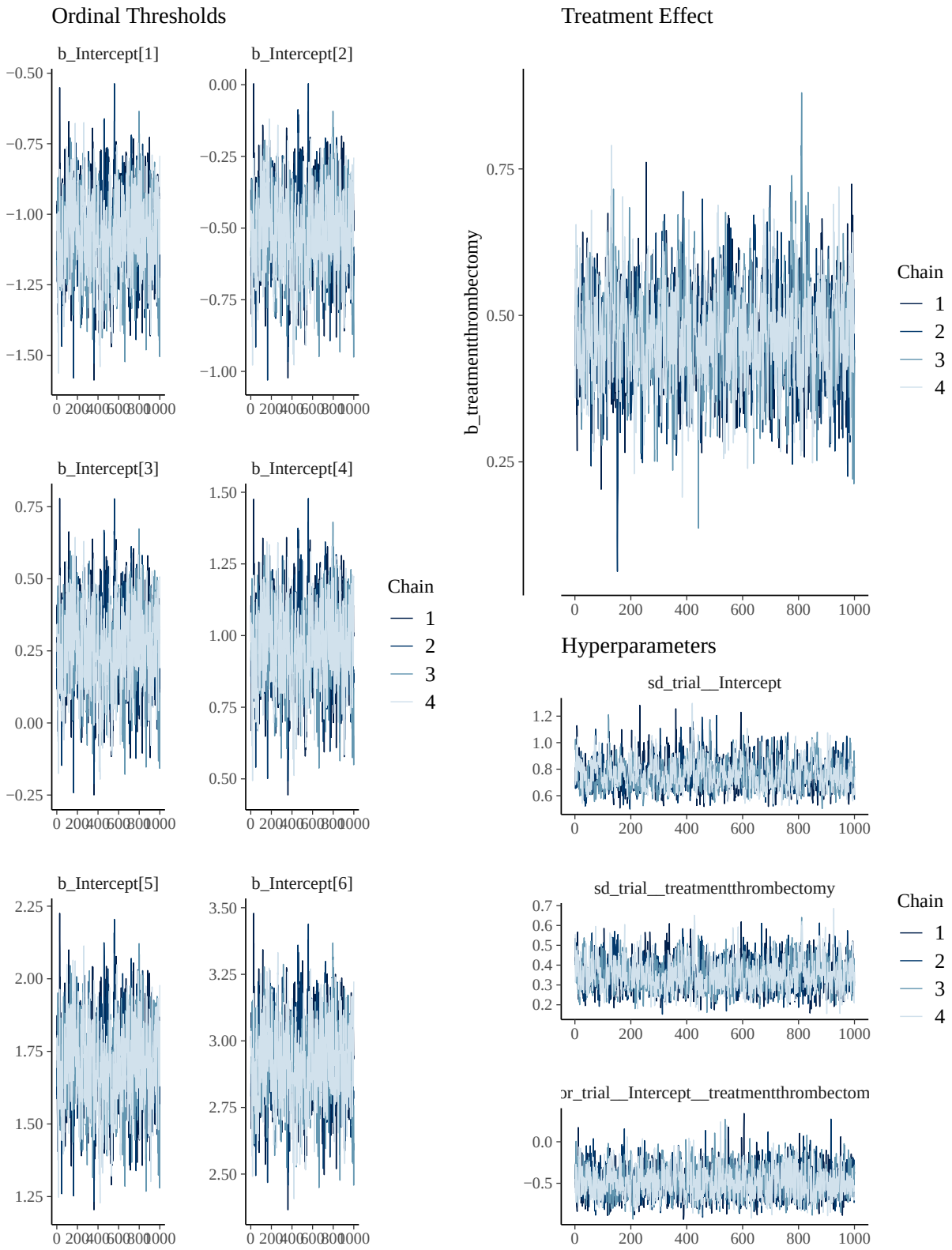

Figure 4: Trace plots for main model parameters.

R-hat and Effective Sample Size

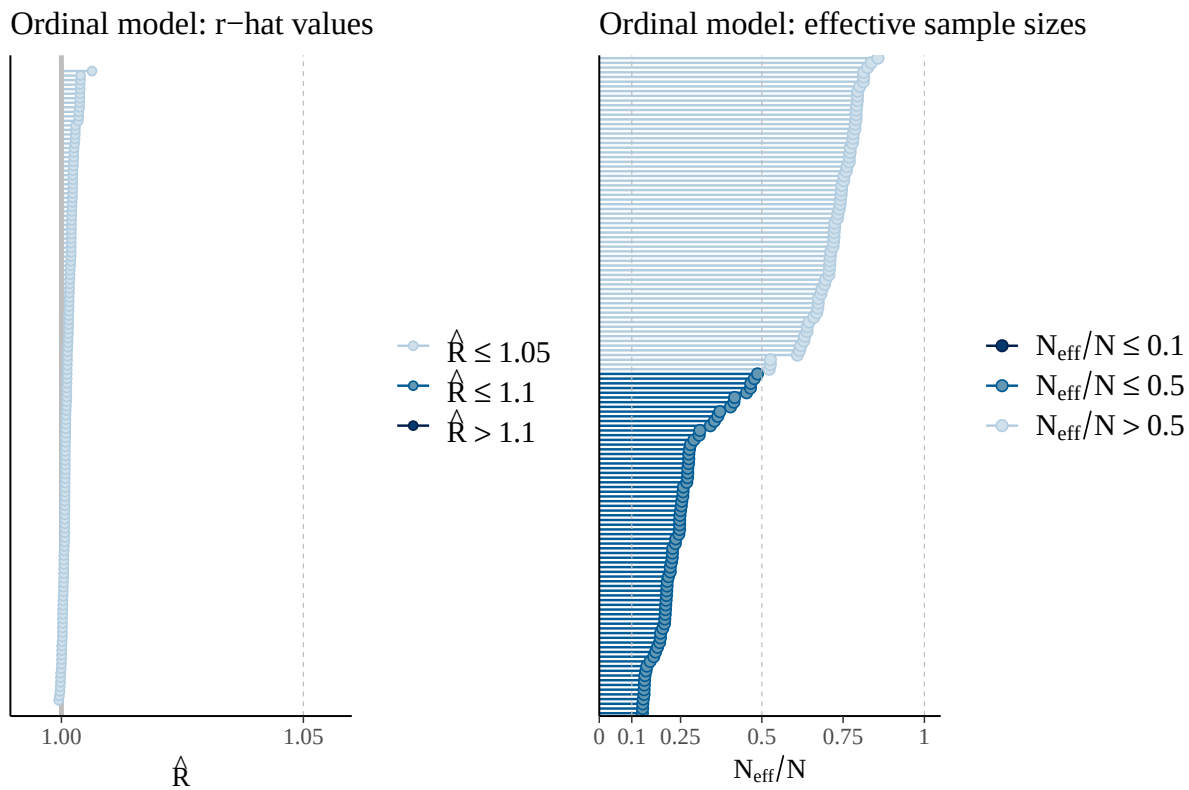

Figure 5: R-hat values and effective sample sizes for main model.

Posterior Predictive Check

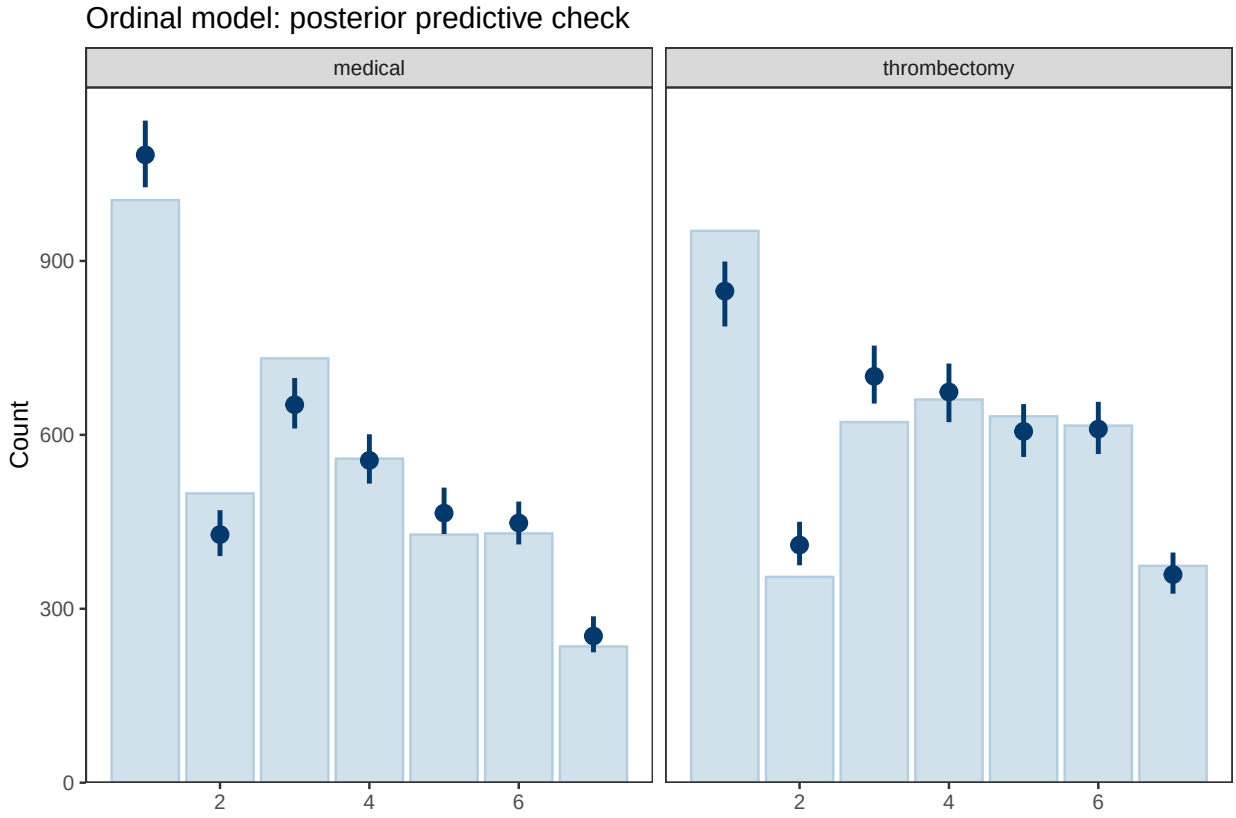

Figure 6: Posterior predictive check for the main model, grouped by treatment arm.

Prior and Likelihood Sensitivity

Prior power-scaling sensitivity analysis for key model parameters. Sensitivity was low across all parameters.

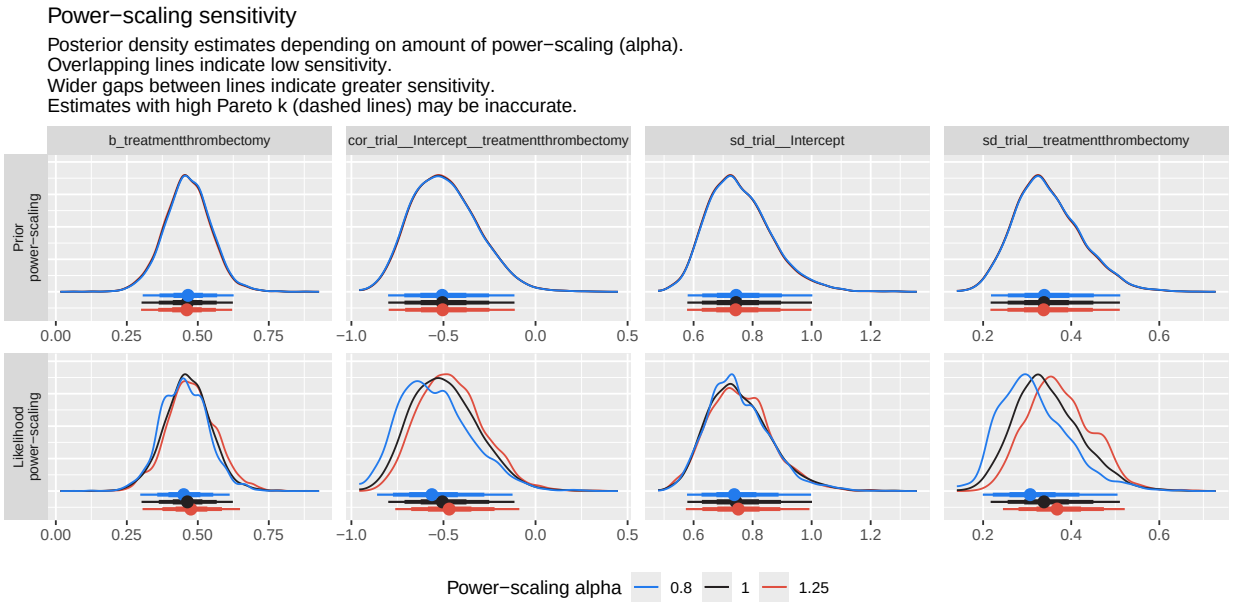

Figure 7: Prior sensitivity (power-scaling) analysis for the main model.